

1 Answers in II, III, and IV

II.

1.) Find the (a) first, (b) second, (c) third, and (d) fourth moments of the set 2, 3, 7, 8, 10, 10, 12, 15, 17, 20.

1.0.1 Solution:

Dataset:

2, 3, 7, 8, 10, 10, 12, 15, 17, 20

Sample size:

$$n = 10$$

Mean:

$$\bar{x} = \frac{2 + 3 + 7 + 8 + 10 + 10 + 12 + 15 + 17 + 20}{10} = \frac{104}{10} = 10.4$$

a) First Moment

$$m'_1 = \frac{\sum x}{n} = \frac{104}{10} = 10.4$$

b) Second Moment

$$\begin{aligned} m'_2 &= \frac{\sum x^2}{n} \\ &= \frac{4 + 9 + 49 + 64 + 100 + 100 + 144 + 225 + 289 + 400}{10} \\ &= \frac{1384}{10} = 138.4 \end{aligned}$$

c) Third Moment

$$\begin{aligned} m'_3 &= \frac{\sum x^3}{n} \\ &= \frac{8 + 27 + 343 + 512 + 1000 + 1000 + 1728 + 3375 + 4913 + 8000}{10} \\ &= \frac{20906}{10} = 2090.6 \end{aligned}$$

d) Fourth Moment

$$\begin{aligned} m'_4 &= \frac{\sum x^4}{n} \\ &= \frac{16 + 81 + 2401 + 4096 + 10000 + 10000 + 20736 + 50625 + 83521 + 160000}{10} \\ &= \frac{341476}{10} = 34147.6 \end{aligned}$$

2. Find the (a) first, (b) second, (c) third, and (d) fourth moments about the mean of the set 2, 3, 7, 8, 10, 10, 12, 15, 17, 20.

Dataset:

2, 3, 7, 8, 10, 10, 12, 15, 17, 20

Mean:

$$\bar{x} = 10.4$$

Formula:

$$m_r = \frac{1}{n} \sum (x - \bar{x})^r$$

a) First moment about the mean

$$m_1 = 0$$

b) Second moment

$$m_2 = 30.24$$

c) Third moment

$$m_3 = 33.352$$

d) Fourth moment

$$m_4 = 2048.9952$$

3. Verify:

$$\begin{aligned} m_4 &= m'_4 - 4m'_1m'_3 + 6(m'_1)^2m'_2 - 3(m'_1)^4 \\ m_4 &= 34147.6 - 4(10.4)(2090.6) + 6(10.4)^2(138.4) - 3(10.4)^4 \\ &= 34147.6 - 86968.96 + 89937.2544 - 35166.2592 \\ &= 2048.9952 \end{aligned}$$

III

Prove that

$$m'_4 = m_4 + 4hm_3 + 6h^2m_2 + h^4$$

where

$$h = m'_1$$

Substituting the values:

$$\begin{aligned} m'_4 &= 2048.9952 + 4(10.4)(33.552) + 6(10.4)^2(30.24) + (10.4)^4 \\ &= 2048.9952 + 1395.7632 + 19655.9016 + 11448.5856 \\ &= 34147.6 \end{aligned}$$

IV.

1. The area between $-t_1$ and t_1 is 0.90 and $v = 20$.

$$1 - 0.90 = 0.10$$

$$\frac{0.10}{2} = 0.05$$

So,

$$t_{0.05,20}$$

From the t -table,

$$t_1 \approx 1.725$$

2. The area to the left of $-t_1$ is 0.025 and $v = 20$.

This also means the right tail area is also 0.025.

So,

$$t_{0.025,20}$$

From the t -table,

$$t_1 \approx 2.086$$

3. The combined area to the right of t_1 and left of $-t_1$ is 0.01 and $v = 5$.

Total area:

$$0.01$$

Each tail:

$$\frac{0.01}{2} = 0.005$$

So,

$$t_{0.005,5}$$

From the t -table,

$$t_1 \approx 4.032$$

4. The area to the right of t_1 is 0.55 and $v = 16$.

Right tail probability:

$$P(T > t_1) = 0.55$$

So,

$$P(T < t_1) = 0.45$$

and using the t -distribution,

$$t_1 \approx -0.13$$